SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: CSA0703

CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB = 150 × 10^6 = 150,000,000 bytes**
- Segment size = **1500 bytes**
- Data Link overhead/frame = **40 bytes**
- Bandwidth = **50 Mbps**

(i) Number of segments

$$\frac{150{,}000{,}000}{1500} = 100{,}000 \text{ segments}$$

Since each segment becomes one frame,

Frames transmitted = 100,000

(ii) Total overhead

$$100{,}000 \times 40 = 4{,}000{,}000 \text{ bytes}$$

= **4 MB**

(iii) Total data transmitted

$$150 + 4 = 154 \text{ MB}$$

Convert to bits:

$$154 \times 8 = 1232 \text{ Mb}$$

Transmission time

$$\frac{1232}{50} = 24.64 \text{ s}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Overhead = **4 MB**
- Transmission time ≈ **24.64 seconds**

Explanation

- **Transport Layer:** Segments data, performs sequencing, acknowledgments, retransmission, and flow control.
- **Data Link Layer:** Adds frame headers/trailers, performs error detection (CRC), and ensures reliable node-to-node delivery.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = 6
- Total frames = 48

(i) Number of windows

$$\frac{48}{6} = 8$$

(ii) Retransmissions

One frame lost per window.

$$8 \times 1 = 8$$

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents sender overflow by allowing only a limited number of outstanding frames, improving network efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- Packet = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Maximum payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Number of fragments

$$\frac{12000}{1480} = 8.1$$

Need 9 fragments

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = 9
- Header overhead = 180 bytes
-

Explanation

The Network Layer fragments packets, routes each fragment independently, and the destination reassembles them using fragment offsets and identification fields.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer

- Number of windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents congestion and allows efficient utilization of bandwidth.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Number of checkpoints before failure

Checkpoints at

60,120,180,240,300,360

Total = **6 checkpoints**

Data to retransmit

Last checkpoint = 360 MB

Failure = 390 MB

Retransmit

$$390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation

Synchronization checkpoints allow communication to resume from the last checkpoint instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Percentage reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation

Presentation Layer performs compression, encryption, and format conversion, improving efficiency and security.

7.An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

Number of requests

$$\frac{3000}{3} = 1000$$

Transmission time

Convert data into Mb

$$3000 \times 8 = 24000 \text{ Mb}$$
$$\frac{24000}{120} = 200 \text{ seconds}$$

Answer

- Requests = **1000**
- Transmission time = **200 seconds**

Explanation

The Application Layer provides services such as FTP, file management, authentication, and error reporting.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Processing delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Propagation delay = 18 ms

Transmission delay = 12 ms

Total delay

$$15 + 18 + 12 = 45 \text{ ms}$$

Answer

- Total delay = **45 ms**

Explanation

- Application: User services
- Transport: End-to-end delivery
- Network: Routing
- Data Link: Framing
- Physical: Signal transmission

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame = 1600 bytes
- Overhead = 80 bytes

Useful payload

$$1600 - 80 = 1520 \text{ bytes}$$

Frames required

$$\frac{80{,}000{,}000}{1520} = 52{,}631.58$$

Need **52,632 frames**

Total overhead

$$52{,}632 \times 80 = 4{,}210{,}560 \text{ bytes}$$

≈ **4.21 MB**

Transmission efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**
- Overhead = **4,210,560 bytes (≈4.21 MB)**
- Efficiency = **95%**

Explanation

Protocol overhead reduces effective throughput because extra bytes consume bandwidth without carrying user data.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Frame overhead = 60 bytes

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

= 168,000,000 bytes

Number of segments

$$\frac{168{,}000{,}000}{1400} = 120{,}000$$

Protocol overhead

$$120{,}000 \times 60 = 7{,}200{,}000 \text{ bytes}$$

= 7.2 MB

Final transmitted data

$$168 + 7.2 = 175.2 \text{ MB}$$

Answer

- Compressed file size = 168 MB
- Segments = 120,000
- Protocol overhead = 7.2 MB
- Final transmitted data = 175.2 MB

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data and provides reliable delivery.
- **Data Link Layer:** Adds framing and error detection**.**
- **Physical Layer:** Transmits the final bit stream over the communication medium**.**